

N01K-LA

Product Specifications

Version 1.0 Date 2024.12

illustrate

This document corresponds to the N01K-LA **product specification** .

This document is intended for system engineers, development engineers, and test engineers.

Our company will not be held responsible for any personal injury or property loss caused by improper operation of customers .

Due to product version upgrades or other reasons, the contents of this document will be updated as necessary without prior notice.

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About this document

scope

This document corresponds to the N01K-LA **product specification** .

Target Audience

This document is intended for system engineers, development engineers, and test engineers.

Revision History

Version	date	change	Version
1.0	2024-01	Initial release	V1.0.0

Symbol Conventions

symbol	meaning
	Danger or warning, rules that users must follow, otherwise it may cause irreversible damage to the module or customer equipment, or even personal injury.
	Caution warns users of areas where they should pay special attention when using the module. If they fail to comply, the module or customer equipment may malfunction.
	Instructions or tips, providing opinions or suggestions on using the module.

1 Product Introduction

1.1 Product Name

Product Name : N01K-LA Wired Equipment

Product Model : N01K-LA

Product specifications : 80 mm* 37 mm*1 5.4 mm

1.2 Product appearance



1.3 Basic parameters of the equipment

1.3.1 Machine parameters

Product Parameters	describe
Hardware Platform	GX316 + AT6558R/HD8122
Machine size	80 mm* 37 mm* 15.4 mm
GPS Antenna	Built-in 25 mm* 25 mm* 4 mm ceramic antenna
Positioning method	GPS/BDS/GLONASS & WIFI & LBS
Power Input	9~90VDC
Shock Sensor	support
Battery Installation	welding
Battery capacity	300mAh
Cut off oil and power	support
Power Status Indicator	red
GPS Status Indicator	blue
Network status indicator	green
ACC ignition detection	support
Working Mode	Long Connection
SMS	VOLTE SMS
Phone Features	Not supported
LTE Antenna	Shrapnel Antenna
Communication Network	CAT1
Communication	TCP
Operating temperature	-20 °C ~+ 70 °C
Storage temperature	-25 °C~+ 75 °C optimal (0 °C~ + 30 °C)

Positioning error	GPS positioning error ≤ 10 M (the actual positioning error is related to factors such as the vehicle's regional environment and weather). The base station positioning error depends on the current base station coverage information. The WIFI positioning error depends on the accuracy of the third-party database.
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1.3.2 Network performance

Support Mode	4G-LTE-CAT1
Frequency band	B1 B 3 B 7 B 8 B 28 F B 66

1.3.3 GPS Performance

Positioning type	GPS/BDS/GLONASS
Signal reception	Support GPS/BDS/GLONASS parallel reception and joint positioning
Cold Start TTF	≤ 35 S
Hot Start TTF	≤ 1 S
Recapture TTF	≤ 1 S
Cold start capture sensitivity	-14.8 dBm
Recapture sensitivity	-1.60 dBm
Tracking sensitivity	-16.2 dBm
Positioning accuracy	≤ 10 M
Speed measurement accuracy	<0.1 m/s (1σ)

Positioning up date rate	Max. 5 Hz
Timing accuracy	<30ns (1 σ)
AGPS	support

2 Functional Description

2.1 Indicator Light Description

Network mode light (yellow light)	Status Definition
Flash	Network signal initialization
Slow Flash	Network signal reception is normal
Not bright	No network signal found or in sleep mode
GPS signal light (blue light)	Status Definition
Flash	Searching for GPS signal
Slow Flash	GPS Positioned
Not bright	No GPS signal found or in sleep mode
POWER light (red light)	Status Definition
Slow Flash	Normal operation with single battery power supply
Always on	External power supply works normally
Not bright	Device failure or hibernation

2.2 Functional Description

Function details	Functional Description
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Speeding alarm	When the speed exceeds the set threshold, an overspeed alarm is uploaded. Default: Off;
Electronic fence	When the terminal exceeds the set fence, a fence alarm will be generated . Default: Off;
Vibration alarm	The default setting of vibration alarm is off. The steps to set the vibration alarm are as follows: Turn on the vibration alarm → The device is in the static state (ACC OFF) , and after 10 minutes, the device enters the defense state → vibrates 3 times within 10 seconds → delay 1 minute → upload the vibration alarm!
Displacement alarm	Turn on the displacement alarm → the device is stationary (ACC OFF) and enters sleep mode for 20 minutes → drives again to the set displacement distance → generates a displacement alarm 2. When the vehicle moves a set distance, the platform receives a displacement alarm. If the alarm mode is selected with SMS alarm and an SOS number is set, the SOS number will receive the corresponding SMS or phone alarm of the displacement alarm.
Low battery alarm	The default setting for low battery alarm is: On, disconnect the external power for 5 minutes, and then upload the low battery alarm .
Disconnect/restore oil and electricity	ACC ON or OFF state, the device is in positioning state, the vehicle speed is less than 20KM/H, if the vehicle speed is greater than 20KM/H, it will be delayed until the conditions are met! 1、 Disconnect oil and electricity: Send command on the platform: Disconnect oil and electricity or SMS command: RELAY, 1# 2. Restore oil and electricity: Send command on the platform: Restore oil and electricity or SMS command: RELAY,0#
Power failure alarm	After the device is disconnected from the external power supply, it will send out a power failure alarm within 30 seconds.

Note: If the battery voltage is above 3.8V, the switch must be turned on.
